Attention-Based Multiscale Feature Fusion for Efficient Surface Defect Detection

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***Abstract*— Deep-learning-based detection methods have been widely applied to industrial defect inspection. However, directly using vanilla detection methods fails to achieve satisfying performance due to the lack of identifiable features. In this article, a novel attention-based multiscale feature fusion module (AMFF) is proposed, aiming to enhance defect features and improve defect identification by leveraging attention mechanism in the feature fusion. AMFF includes self-enhanced attention module (SEAM) and cross-enhanced attention module (CEAM). SEAM is performed on a single feature map, which first adopts multiple dilation convolutions to enrich contextual information without compromising resolution and then utilizes the intralayer attention on the current feature map. CEAM takes both the current feature map and the adjacent feature map as input to perform cross-layer attention. The adjacent feature map is modulated with the guidance of the current feature map, which is then combined with the current feature map and the output of SEAM for final prediction. AMFF is utilized in current feature fusion networks, e.g., feature pyramid network (FPN) and path aggregation FPN (PAFPN), and is further integrated into prevalent detectors to guide them to pay more attention to defects rather than the background. Extensive experiments are conducted on two real industrial datasets released by Tianchi platform, i.e., fabric and aluminum defect datasets. For each dataset, 500 images are randomly selected for test and the rest for training. The proposed AMFF is demonstrated to significantly boost defect detection accuracy with acceptable computational cost, and the real-time performance could fully satisfy practical requirements.**

Manuscript received 24 September 2023; revised 16 January 2024; accepted 10 February 2024. Date of publication 7 March 2024; date of current version 14 March 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2021ZD0114505, in part by the Major Basic Research Projects of Natural Science Foundation of Shandong Province under Grant ZR2019ZD07, and in part by the National Natural Science Foundation of China under Grant 62303321. The Associate Editor coordinating the review process was Dr. Shovan Barma. *(Yuhao Zhao, Qing Liu, and Hu Su contributed equally to this work.) (Corresponding authors: Wei Zou; Song Liu.)*

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Digital Object Identifier 10.1109/TIM.2024.3372229

***Index Terms*— Attention mechanism, feature enhancement, feature fusion, object detection, surface defect detection.**

I. Introduction

SURFACE defect detection [1] plays a pivotal role in industrial applications, particularly in the realm of quality control. Manual defect detection is labor-intensive, time-consuming, and sometimes causes secondary damage, which does not satisfy the growing demand of industrial applications. Additionally, human subjectivity will inevitably affect the consistency of product quality, especially for scenes with weak visual effects. Vision-based defect detection provides a promising solution to deal with the difficulties due to its high efficiency and reliable performance.

Traditional vision-based defect detection methods [2], [3] mainly rely on task-specific features extracted by the descriptors to identify and locate defect regions. The descriptors are meticulously crafted to extract specific defect attributes such as edge [4], contour [5], and texture [6], making it challenging to attain comprehensive representation of defects. The advent of convolutional neural network (CNN) accelerates the development of computer vision by learning features automatically. Specially, CNN-based object detection methods, notably pioneered by Faster R-CNN [7], you only look once (YOLO) [8], and RetinaNet [9], have achieved state-of-the-art (SOTA) performance in various detection tasks [10], [11], [12], [13] and are widely used in industrial applications [14], [15], [16]. Some previous researchers focused primarily on directly applying the well-demonstrated detectors to defect detection, while the characteristics of defects and their differences with common objects are neglected. In the following, limited efforts [17], [18] have been devoted to further improvement. However, challenging issues remain in the field of defect detection.

Defects are usually weak and nonsalient, as shown in Fig. 1. Typical fabric defects and their zoom-in views are shown in Fig. 1(a). The defects both exhibit indistinct boundaries and low contrast with the background, in which the disparity in gray-scale values along the edges is within a 10-pixel range. The feature information of defects can be easily overwhelmed by background noise. In addition, for defects such as scratches, the long, thin morphology makes them occupy only a minute fraction of pixels within their enclosing rectangle bounding box. Accordingly, defect pixels support weak responses during convolution in the feature extraction process, leading to indistinguishable defect features. This inherent challenge significantly hampers the accurate localization and classifi-

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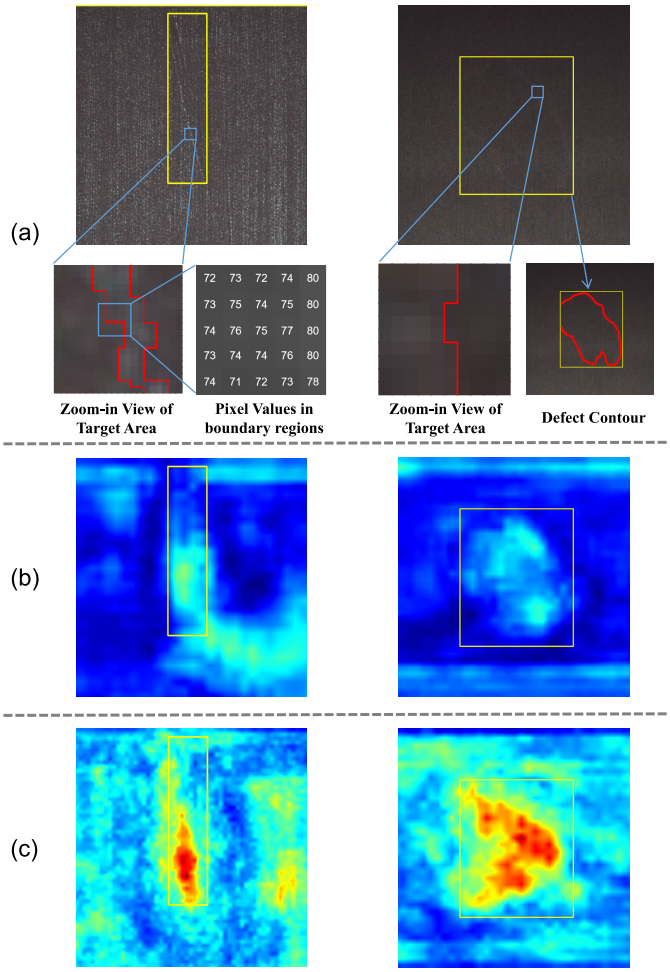


Fig. 1. Typical fabric defects and visualization of the feature maps. (a) Shows typical fabric defects, i.e., the scratch defect (left) and the oil-stained defect (right). The zoom-in views of selected areas are provided to illustrate the details of the junction of the foreground and background. The edges or contours of the defects are marked in red in the zoom-in views. (b) and (c) Visualize the feature maps in which the color represents the degree of activation. With higher activation, features become more distinguishable. The feature maps obtained by FPN and the proposed AMFF are shown in (b) and (c), respectively.

cation of defect regions, and thus degrades the performance of defect detection. This article focuses on improving the representation and identification of defect features for efficient defect detection.

Feature fusion network [19] enriches multiscale information to improve feature representation. Mainstream feature fusion methods commonly use direct addition [19], [20], weighted addition [21], or concatenation [22] to fuse multilevel feature maps, which are insufficient for improving the representation of defect features. Examples are provided in Fig. 1(b) which shows the feature maps of fabric defects derived by feature pyramid network (FPN) [19]. As can be observed from the figure, it is challenging to evidently identify the defects from the background by using FPN features.

Attention mechanism provides an efficient means to enhance object features by forcing the network to pay more attention to object features rather than the background. In this article, we attempt to incorporate attention into feature fusion network to improve feature representation as well as identification. An attention-based multiscale feature fusion module (AMFF) is proposed, which consists of two key branches: a self-enhanced attention module (SEAM) branch and a cross-enhanced attention module (CEAM) branch. In the SEAM branch, dilated convolution [23] is conducted to enrich contextual information without loss of resolution, following an intralayer attention mechanism that consists of channel attention (CA) and spatial attention (SA). The CEAM branch refines the adjacent feature map by adjusting the pixel-wise feature weights, which is guided by the current feature map through a cross-layer attention mechanism. Then, the multiscale features generated by the two branches, along with an additional residual module, are added for the final prediction. The enhanced AMFF features are visualized in Fig. 1(c). With the comparison shown in Fig. 1(b) and (c), the AMFF exhibits the improvement on the identification of defect features. Extensive experiments demonstrate that AMFF can be integrated into popular detectors and bring significant performance improvements.

The contribution of the article is summarized as follows. First, we point out feature representation and identification problems in defect detection, for which the AMFF is constructed by incorporating attention mechanism into the feature fusion networks. Second, AMFF enriches the feature maps with contextual information, and performs intralayer and cross-layer attention simultaneously to improve defect feature representation and identification, and thus defect detection performance. Finally, the proposed AMFF can be seamlessly integrated into the feature fusion networks and applied to prevalent detectors. Extensive experiments demonstrate that AMFF significantly boosts the performance of defect detection.

The rest of this article is organized as follows. In Section II, we review the related works of deep-learning-based defect detection, feature fusion networks in object detection, and attention mechanism in CNN. In Section III, the overviews of the defect detection network and the proposed AMFF are elaborated. In Section IV, we present ablation and comparison experiments to demonstrate the effectiveness of the proposed method. Finally, the conclusion is drawn in Section V.

II. Related Work

A. Deep-Learning-Based Defect Detection

The advent of deep learning technique promotes the advances of object detection by enabling automatic feature extraction, which greatly improves the generalization across different application scenarios. Significant progress has been made by directly applying the vanilla detectors to industrial defect detection. Faster R-CNN is applied to detect concrete cracks and steel corrosion [24], surface defects of polymeric polarizer [25], and sewer pipe defects [26]. The detector, YOLO, is used to detect surface defects on steel strips [27]. A cascade method is introduced in [28] for metallic surface defect detection. Notably, aforementioned methods commonly choose ResNet [29] as the backbone for feature extraction due to its simplicity and excellent performance.

Despite the promising results, recent studies aim to further improve the performance by exploring defect characteristics.

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Considering the consistency and regularity of the background in fabric images, Mei et al. [17] proposed a convolutional denoising autoencoder network (CDAE) to locate irregular fabric textures, which uses only defect-free samples for training. Tu et al. [18] proposed a cascaded defect detection network to resolve the issue of erroneous detection stemming from the resemblance between various classes of rail defects. The approach involves the initial employment of a segmentation network to precisely locate rail defects, followed by a classification network that achieves fine categorization of rail defects by leveraging high-resolution rail images. However, the challenge of unidentifiable features resulting from weak defects has not been adequately addressed by existing methods, which leads to defect misdetection and consequently degrades defect detection performance. Focusing on this challenge, this article proposes the AMFF to enhance defect representation, and thus the detection performance of weak defects is improved.

B. Feature Fusion Network in Object Detection

Feature fusion is a key concern in object detection, which aims to enhance feature representation with the fusion of multiscale features. Early methods primarily focus on single feature map enhancement either by employing multiple dilated convolutions with different rates [30] or by the aggregation of multilevel feature maps from different convolutional layers [31]. However, the hierarchical feature maps derived by the CNN model have not been fully explored in the detection. FPN [19] constructs a top-down pathway to realize the fusion from high-level semantic information to low-level location information and addresses multiscale detection in a divide and conquer manner. Following works further enhance the fusion by constructing additional pathways traversing through the top and bottom feature maps in a direct addition [20] or weighted addition [21] manner. The above methods effectively realize multiscale feature fusion, without making a distinction between foreground object features and background or noise features. If the object features are relatively weak, those of background or noise would be overwhelmed. As a result, the object features cannot be availably identified. In order to effectively increase the weights of object features in the fusion, this article incorporates the attention mechanism into feature fusion to force the network to pay more attention to useful defect features rather than the background, making the defects distinguishable and improving defect identification.

C. Attention Mechanism in CNN

Attention mechanism helps the model focus on useful features while suppressing others, which effectively enhances feature identification and thus improves identification accuracy. Hu et al. [32] introduced CA mechanism to adaptively adjust the relative proportion of different channels, and Woo et al. [33] decoupled CA and SA in a more lightweight attention module. Feature-aligned pyramid network (FaPN) [34] proposes the feature selection module to explicitly model the importance of channel responses of multilevel feature maps and recalibrate them accordingly. The above attention modules are all applied to one single-layer feature map to perform intralayer attention in either feature extraction or feature fusion manner. A multiscale CA module is proposed in [35] to learn the fusion weights of multilevel feature maps by taking the sum of them as input, which generalizes the attention-based feature fusion to cross-layer scenarios. An adjacent context coordination network (ACCoNet) [36] is proposed to coordinate adjacent contextual features by conducting SA mechanism on adjacent layers, where the correspondence between channels in multiple feature maps is ignored. As can be concluded, the above methods have not taken full advantage of attention mechanism to construct an efficient multiscale feature fusion network. In this article, we incorporate both intralayer attention and cross-layer attention in feature fusion to construct the AMFF. AMFF simultaneously employs CA and SA with sufficient contextual information to enhance feature identification. The proposed method could be applied to prevalent detectors to improve defect detection performance with improved defect representation.

III. Method

In this section, a network overview is described in Section III-A, following which the proposed attention-based multiscale feature fusion module is described in Section III-B.

A. Overview

1) Overview of Defect Detection Network: As shown in Fig. 2, overview of defect detection network (DDN) based on deep learning consists of three parts: backbone for feature extraction, neck for feature fusion, and head for prediction. CNNs are the basis of DDN and are used in all three parts of the network. Taking images with size of $H \times W \times 3$ as input, the backbone network extracts multiscale features through multilayer convolutions, such as low-level edge and texture features and high-level semantic features. Commonly used backbone networks include ResNet [29], Visual Geometry Group (VGG) [37], and MobileNet [38], which are usually pretrained on large-scale image data and then fine-tuned for defect detection tasks. In this article, ResNet is chosen as the backbone to extract multiscale features, which consists of five blocks. And the feature maps of the last four blocks are adopted, which are denoted as $C_i \in \mathbf{R}^{h_i \times w_i \times c_i}$ $(c_{i=\{2,3,4,5\}} = \{256, 512, 1024, 2048\})$, for subsequent processing. And h_i and w_i can be formulated as $H/2^i$ and $W/2^i$.

The neck network is an important part for integrating features from multilayer feature maps. Prediction with single-layer feature maps has been proven to be inadequate, which unable to adapt to the detection of objects with different scales. To make better use of multilayer feature maps, the pyramid structure has been introduced to achieve the purposes of feature fusion and feature reuse.

a) FPN [19]: is a classical pyramid network that fuses the features of different layers in a top-down manner. As shown in Fig. 2, taking feature maps C_i and C_{i+1} as input, FPN first adopts a 1×1 convolution and upsampling to align the channel and size of feature maps in different layers,

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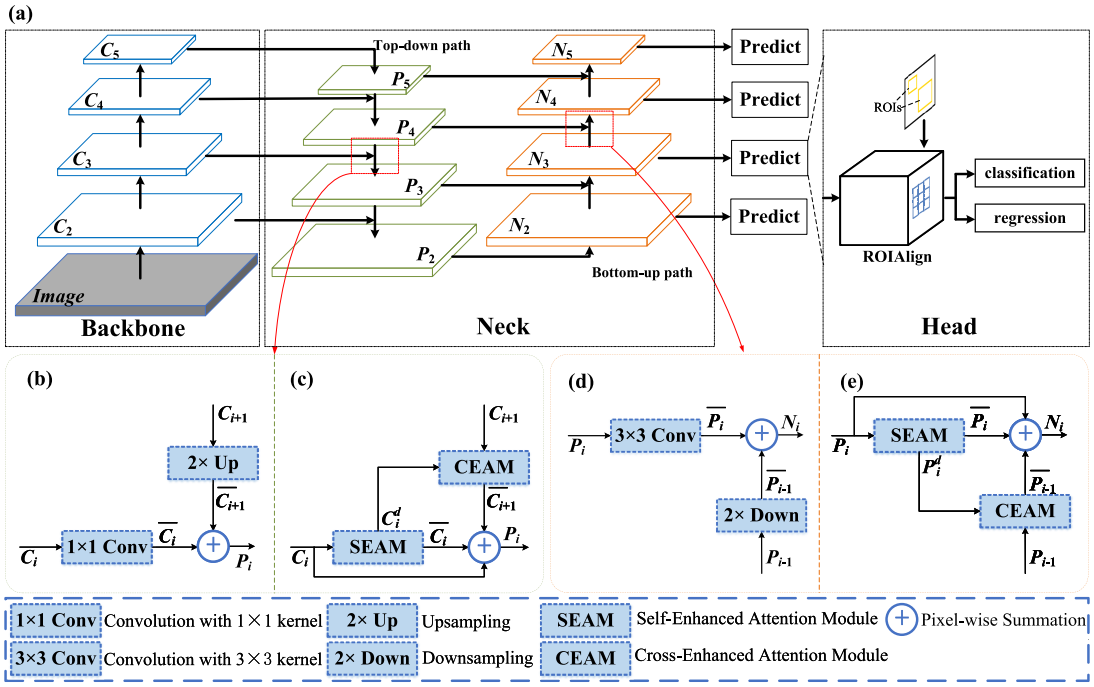


Fig. 2. Overview of the framework. (a) DDN. In a single pass, original feature maps from different stages of the backbone network (ResNet-101) are employed to generate feature pyramid. Multiscale feature maps are fused by the neck network via a top-down path (FPN) and an additional bottom-up path (PAFPN). The detailed structures of FPN and PAFPN are illustrated in (b) and (d), and those of the corresponding AMFF and AMFF* are shown in (c) and (e). The feature maps output by the neck are used for classification and regression in the following head.

respectively. Then, adjacent feature maps are fused by a simple "add" operation. The whole process can be formulated as

$$P_i = \text{Conv}_{1*1}(C_i) + \text{Up}(C_{i+1}), \quad i = 2, 3, 4 \tag{1}$$

where $P_i \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of FPN, $\text{Conv}_{1*1}(\cdot)$ is a standard convolutional layer with 1×1 kernel, and $\text{Up}(\cdot)$ is the $2\times$ upsampling operation implemented by bilinear interpolation.

b) Path Aggregation FPN (PAFPN) [20]: serves as an extension of the FPN. A bottom-up path is additionally adopted to reprocess the output of the FPN, which can be defined as

$$N_i = \text{Conv}_{3*3}(P_i) + \text{Down}(P_{i-1}), \quad i = 3, 4, 5 \tag{2}$$

where $N_i \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of PAFPN, $\text{Conv}_{3*3}(\cdot)$ is a standard convolutional layer with 3×3 kernel, and $\text{Down}(\cdot)$ is the $2\times$ downsampling operation implemented by maxpooling. We can flexibly choose one of the above two fusion methods as our neck. Through these operations, multiscale features can be effectively fused, resulting in enriched information within a single feature map.

The head network is the final part of DDN, which is responsible for predicting the category and location of objects. In this part, regions of interest (ROIs) are clipped by proposals generated by region proposal network (RPN) in two-stage detectors or by anchors generated in single-stage detectors. Then, the ROIs are transformed into fixed-size feature vectors through the ROI Align operation to predict a one-of-(C+1) defect category and a refined bounding box by classification and detection heads, respectively.

2) Attention-Based Multiscale Feature Fusion: In practice, as shown in Fig. 2(c), AMFF is composed of two branches: a SEAM branch and a CEAM branch. In the SEAM branch, multiple dilated convolutions with different dilation rates are used in parallel to enhance local and global information simultaneously, which can effectively obtain defect features of different shapes and sizes. Additionally, SEAM utilizes the intralayer attention mechanism to adaptively adjust the weights of different features according to their effectiveness. The CEAM branch takes the current and adjacent feature maps as input, and the cross-correlation information between them is found by a cross-layer attention mechanism. The adaptive fusion mechanism between the current layer and its adjacent layer enables the proposed AMFF to transmit valuable information to the subsequent head network. AMFF could be applied to the neck part of DDN, such as FPN or PAFPN, to enhance defect representation in the feature fusion, as Fig. 2 shows. Details of AMFF are presented in Section III-B, and its effectiveness is demonstrated in Sections IV-B and IV-C.

B. Attention-Based Multiscale Feature Fusion Module (AMFF)

Taking FPN as the baseline, we use AMFF to improve the fusion mode in the top-down path which is the key of FPN. It focuses on feature adjustment between the current layer and its adjacent layer, which is illustrated in Fig. 3 in detail. We denote the transformation accomplished by AMFF as $F(\cdot)$, then AMFF can be formulated as follows:

$$P_i = F(C_i, C_{i+1}), \quad i = 2, 3, 4 \tag{3}$$

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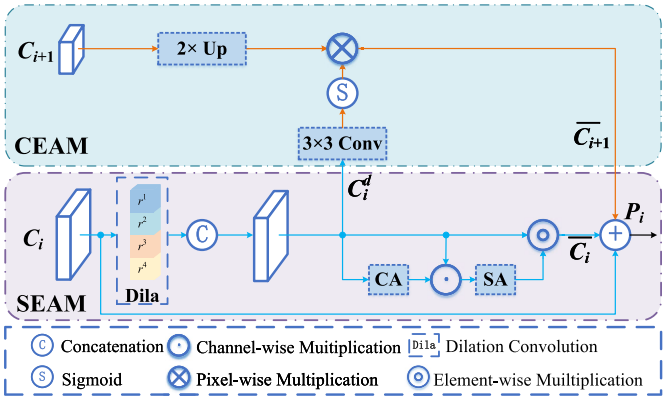


Fig. 3. Structure of the SEAM (bottom) and CEAM (upper) branches, which are indicated by blue and orange arrows, respectively.

where $P_i \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of AMFF. C_i and C_{i+1} are the inputs of AMFF, which represent the feature maps of the ith layer and the $(i+1)$th layer, respectively.

1) SEAM Branch: The SEAM branch implements two main operations on ith layer feature maps. First, several dilated convolutions with different dilation rates are applied in parallel on the ith layer feature maps, which are defined as

$$C_{i,t}^d = \text{Dconv}_{3*3}\left(C_i; r^t\right), \quad t \in \{1, 2, 3, 4\} \tag{4}$$

where $\text{Dconv}_{3*3}(*;*)$ is dilated convolution with 3×3 kernel, $r^t = t$ is dilation rate, and $C_{i,t}^d \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of dilated convolution with dilation rate r^t. This can effectively extract feature information from different receptive fields.

Then, features from the above dilated convolutions are concatenated in parallel followed by convolution operation, so the output $C_i^d \in \mathbf{R}^{h_i \times w_i \times c_i}$ can be defined as

$$C_i^d = \text{Conv}_{3*3}\left(\text{Concat}\left(C_1^d, C_{i,2}^d, C_{i,3}^d, C_{i,4}^d\right)\right) \tag{5}$$

where $\text{Concat}(\cdot)$ is the cross-channel concatenation. This can effectively integrate features from different receptive fields. The subsequent operations in AMFF are based on C_i^d.

In order to force the network to pay more attention to the features of object regions, intralayer attention mechanism, including CA and SA [33], is adopted to suppress redundant information from C_i^d in an adaptive manner, which is formulated as

$$\overline{C}_i = \text{SA}\left(\text{CA}\left(C_i^d\right) \odot C_i^d\right) \odot C_i^d \tag{6}$$

where $\overline{C}_i \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of the SEAM branch, $\odot$ is channel-wise multiplication, and $\odot$ is element-wise multiplication. Specifically, $\text{CA}(\cdot)$ is implemented with a spatial-wise global max pooling (GMP), a 1×1 convolutional layer with rectified linear unit (ReLU) activation function and a 1×1 convolutional layer with sigmoid activation function. $\text{SA}(\cdot)$ is implemented with a channel-wise GMP and a 7×7 convolutional layer with sigmoid activation function. Such an adaptive adjustment process selects valuable information from C_i^d.

The SEAM branch is used to enhance the discrimination between defects and background. The intralayer attention mechanism can adaptively increase the weights of defect features, and the use of dilated convolutions can provide more effective guidance for this adaptation.

2) CEAM Branch: The CEAM branch aims to seek the intrinsic connections between adjacent layers to adaptively fuse valuable features through cross-layer attention mechanism. It can be computed as

$$\overline{C}_{i+1} = \text{Sig}\left(\text{Conv}_{3*3}\left(C_i^d\right)\right) \otimes \text{Up}(C_{i+1}) \tag{7}$$

where $\overline{C}_{i+1} \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of CEAM branch, $\text{Sig}(\cdot)$ is sigmoid activation function, and $\otimes$ is pixel-wise multiplication. Under the guidance of the ith layer features, this branch provides more valuable contextual interaction information for fusion.

3) Branches Integration: After the above effective operations, the outputs of SEAM and CEAM are integrated with the original ith layer feature maps in a residual manner as

$$P_i = \overline{C}_i \oplus \overline{C}_{i+1} \oplus C_i \tag{8}$$

where $\oplus$ is pixel-wise summation. In summary, the ith layer features are adaptively adjusted by intralayer and cross-layer attention mechanisms.

C. AMFF*

We extend AMFF to apply to the bottom-up path in PAFPN, which is denoted as AMFF*. AMFF* takes P_i and P_{i-1} as input and the transformation can be formulated as

$$N_i = F'(P_i, P_{i-1}), \quad i = 3, 4, 5 \tag{9}$$

where $N_i \in \mathbf{R}^{h_i \times w_i \times c_i}$ is the output of AMFF*. In AMFF*, SEAM branch performs similar to AMFF. SEAM takes P_i as input and the output is obtained with (6) which is denoted as $\overline{P}_i \in \mathbf{R}^{h_i \times w_i \times c_i}$. The output of CEAM is derived as

$$\overline{P}_{i-1} = \text{Sig}\left(\text{Conv}_{3*3}\left(P_i^d\right)\right) \otimes \text{Down}(P_{i-1}) \tag{10}$$

where P_i^d is obtained from P_i with (5). Then, N_i, the output of AMFF*, can be computed as

$$N_i = \overline{P}_i \oplus \overline{P}_{i-1} \oplus P_i. \tag{11}$$

The architecture of AMFF* is similar to that of AMFF illustrated in Fig. 3. In the figure, the current feature map C_i is replaced by P_i and the adjacent feature map C_{i+1} is replaced by P_{i-1}. AMFF* utilizes cross-attention in a bottom-up manner, which integrates high-level semantic information with low-level localization information. With the help of AMFF*, the performance of PAFPN would be further improved.

IV. Experimental Results

A. Experimental Settings

1) Datasets and Evaluation Metrics: We evaluate the effectiveness of the proposed method on fabric [39] and aluminum defect datasets [40], which are both released by Tianchi platform. Fabric defect dataset consists of 5913 images with 9523 annotations of bounding boxes, and the defects are divided into 20 categories. Aluminum defect dataset consists of 4356 images, of which 3005 are defect images with 4078 annotations of bounding boxes, and the defects are divided into ten categories. Following [41], we randomly select 500 images for test and use the rest for training. This

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datasetting is adopted in all the experiments for the sake of fairness.

For a more comprehensive evaluation of detection performance, average precision (AP) with specific threshold is adopted as the metric. The metric considers a predicted box as a true positive if its intersection over union (IoU) value with any ground truth box exceeds the threshold. AP@0.5 and AP@0.75 denote the AP values with the thresholds of 0.5 and 0.75, respectively, which evaluate detection performance under different location accuracy. The average value of AP with specific threshold ranging from 0.5 to 0.95 at intervals of 0.05 is used as the metric, which is denoted as mean average precision (mAP) in the following. We further use AP_s, AP_m, and AP_l to evaluate the mAP performances in the detection of small, medium, and large objects, respectively. The objects that occupy less than 32×32 pixels are denoted as small, those that occupy less than 96×96 pixels are medium, and others are large.

2) Implementation Details: The proposed AMFF is applied to the representative detectors equipped with different necks. The detectors include the two-stage detector Faster R-CNN [7], the one-stage detector RetinaNet [9], and the anchor-free detector FCOS [44]. All the detectors adopt their default settings besides the improvement part. Our method is implemented on MMdetection with four NVIDIA Titan RTX GPUs. ResNet is chosen as the backbone network for the excellent performance in feature extraction by the settlement of gradient vanishment through the residual connections. The necks involved in the experiments are FPN and PAFPN. Specially, ResNet-101 with FPN is set as the default backbone and neck unless explicitly specified. The backbone is pretrained on ImageNet [42] and the parameters in the neck and head are initialized by normal distribution [43]. Stochastic gradient descent algorithm is adopted as the optimizer with the weight decay set to 0.0001 and the momentum to 0.9. With batch size of 16, the initial learning rate is set to 0.01, and it will be degraded by multiplying 0.1 at epoch 24 and 33 out of a total of 36 epochs. In training and testing phases, input images are resized to make their shorter side 800 pixels while keeping their longer side less than or equal to 1333 pixels. Input images are randomly flipped horizontally with a probability of 0.5 for data augmentation.

B. Demonstration of AMFF on Fabric Defect Dataset

1) Ablation Experiments: The proposed AMFF aims to address the limitations of feature fusion in object detection. In the ablation experiments, Faster R-CNN is used as the detector and FPN is chosen as the baseline in the ablation studies, where the "add" operations in FPN are replaced by the proposed AMFF in the top-down path. AMFF includes as follows.

1) SEAM, i.e., which adjusts the current feature map with dilation convolutions and an intralayer attention mechanism.
2) CEAM, i.e., which applies the interaction to the adjacent feature map through a cross-layer attention mechanism.

In SEAM, dilated convolution is a crucial component. A larger dilation rate means more contextual information is introduced. We conduct ablation experiments to verify the effects of different dilation rates on fabric defect dataset. The results are shown in Table I. It can be seen from the table that, along with more dilated convolutions with larger dilation rates, the mAP performance rises at the beginning, and then goes down. The dilation rate setting, $r^1 = 1$, $r^2 = 2$, $r^3 = 3$, and $r^4 = 4$, achieves the highest performance (17.5% mAP) and is adopted in the following experiments on fabric defect detection.

TABLE I
ABLATION ANALYSES ABOUT DILATION RATE r ON FABRIC DEFECT DATASET

dilation rate r	mAP
r^1=1	17.0
r^1=1, r^2=2	17.2
r^1=1, r^2=2, r^3=3	17.3
r^1=1, r^2=2, r^3=3, r^4=4	**17.5**
r^1=1, r^2=2, r^3=3, r^4=4, r^5=5	16.6

TABLE II
ABLATION ANALYSES ON FABRIC DEFECT DATASET. THE FIRST LINE REFERS TO THE BASELINE OF FPN, FOLLOWED BY THE RESULTS SEPARATELY USING SEAM AND CEAM. THE RESULT OF SIMULTANEOUS APPLICATION OF SEAM AND CEAM IS SHOWN IN BOLD IN THE LAST ROW

SEAM	CEAM	mAP	AP_{50}	AP_{75}
-	-	16.1	36.7	11.2
√		16.8	37.9	11.7
	√	16.7	37.8	11.2
√	√	**17.5**	**39.1**	**11.8**

Then, we evaluate the design choices of SEAM and CEAM, which are gradually added to analyze their effects. Table II shows the evaluation results. As shown in the table, the application of SEAM and CEAM could stably boost the performance. SEAM improves the performance from 16.1% mAP to 16.8% mAP and CEAM brings a similar improvement of 0.6% mAP. Meanwhile, AP_{50} and AP_{75} performances are improved simultaneously. By using both of them, the proposed AMFF brings a considerable improvement of 1.4% mAP (16.1% versus 17.5%), which demonstrates the significance of AMFF in feature pyramid enhancement. In addition, we conducted ablation experiments on the location of the feature map that guides the integration of adjacent layer. The guidance in C_i^d shows the best performance due to the multiscale information captured by dilated convolution.

2) Performance Demonstration: Examples of fabric defects are shown in Fig. 1(a). The corresponding feature maps obtained by using FPN and AMFF are visualized in Fig. 1(b) and (c), respectively. Evidently, the activation of defect features achieved by AMFF surpasses that obtained with FPN, which means that our approach can guide the network's attention toward the defect regions. This substantially enhances the identification of defect features, benefiting the classification and regression of defects.

For further demonstration, AMFF is applied to the representative detectors equipped with different necks. Taking FPN and PAFPN as baselines, the results of comparison experiments are provided in Table III. It can be seen from the table that the performances of all the detectors has improved. Compared

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TABLE III
PERFORMANCE OF THE SOTA USING AMFF ASSESSED ON FABRIC DEFECT DATASET

Detector	Method	mAP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Faster R-CNN [7]	FPN	16.1	36.7	11.2	12.2	19.2	17.8
	FPN *w* / AMFF	**17.5(+1.4)**	**39.1**	**11.8**	**15.7**	**18.2**	**23.8**
	PAFPN	18.1	40.2	12.0	16.4	18.4	20.7
	PAFPN *w* / AMFF*	**19.1(+1.0)**	**41.9**	**15.3**	**17.0**	**18.9**	**21.5**
RetinaNet [9]	FPN	15.1	37.6	9.1	10.9	15.0	17.1
	FPN *w* / AMFF	**16.9(+1.8)**	**38.7**	**11.3**	**12.4**	**15.1**	**19.0**
	PAFPN	15.6	37.9	9.0	11.4	15.1	18.7
	PAFPN *w* / AMFF*	**17.7(+2.1)**	**41.1**	**12.4**	**11.9**	**15.9**	**21.4**
FCOS [44]	FPN	15.2	34.0	10.3	13.0	17.4	20.7
	FPN *w* / AMFF	**16.9(+1.7)**	**37.1**	**12.2**	**14.2**	**17.9**	**21.3**
	PAFPN	16.2	36.2	11.6	13.5	15.7	21.3
	PAFPN *w* / AMFF*	**17.8(+1.6)**	**38.0**	**14.3**	**14.2**	**18.5**	**23.3**

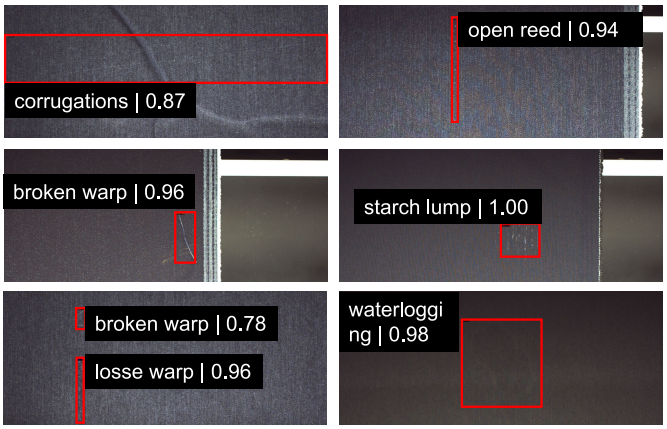


Fig. 4. Examples of detection results on fabric defect dataset.

TABLE IV
ABLATION ANALYSES ABOUT DILATION RATE *r* ON ALUMINUM DEFECT DATASET

dilation rate *r*	mAP
r^1=1	35.2
r^1=1, r^2=2	35.3
r^1=1, r^2=2, r^3=3	35.9
r^1=1, r^2=2, r^3=3, r^4=4	**36.1**
r^1=1, r^2=2, r^3=3, r^4=4, r^5=5	35.9

with FPN, AMFF boosts the performances of the detectors by 1.4% mAP, 1.8% mAP, and 1.7% mAP, respectively. AMFF* improves the performance of the detectors by 1.0% mAP, 2.1% mAP, and 1.6% mAP, respectively, in the comparison with PAFPN. Meanwhile, the performances of AP_{50} and AP_{75} are improved simultaneously. The improvements are obvious, which demonstrate the effectiveness of AMFF. Notably, we observe that PAFPN outperforms FPN on fabric defect dataset possibly due to the additional bottom-up path that shortens the information path between low-layer features and topmost features. Examples of detection results are shown in Fig. 4. The results demonstrate that the proposed AMFF is efficient at detecting weak defects with low contrast. AMFF could obtain accurate prediction results with high confidence.

C. Demonstration of AMFF on Aluminum Defect Dataset

1) Ablation Experiments: We analyze the setting of dilation rate on aluminum defect dataset. The experimental results are shown in Table IV, from which we can see that, the setting, $r^1 = 1$, $r^2 = 2$, $r^3 = 3$, and $r^4 = 4$, achieves the optimal performance and thus is adopted in the following experiments on aluminum defect detection. We further conduct ablation experiments to comprehensively validate the effectiveness of SEAM and CEAM. The results are provided in Table V. As shown in the table, a single SEAM or CEAM could stably boost the performance. The improvement by using SEAM is 1.2% mAP while that of CEAM is 0.8% mAP. Integrating both SEAM and CEAM within the proposed AMFF yields a substantial improvement of 1.4% mAP.

TABLE V
ABLATION ANALYSES ON ALUMINUM DEFECT DATASET. THE FIRST LINE REFERS TO THE BASELINE OF FPN, FOLLOWED BY THE RESULTS SEPARATELY USING SEAM AND CEAM. THE RESULT OF SIMULTANEOUS APPLICATION OF SEAM AND CEAM IS SHOWN IN BOLD IN THE LAST ROW

SEAM	CEAM	mAP	AP_{50}	AP_{75}
-	-	34.7	48.9	36.0
√		35.9	50.8	36.4
	√	35.5	49.7	36.3
√	√	**36.1**	**50.9**	**36.7**

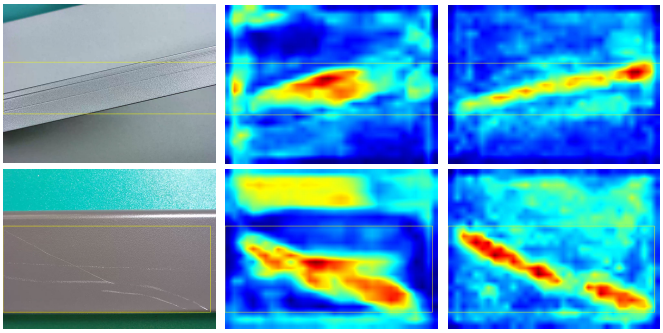


Fig. 5. Feature maps of aluminium defects. (a) Presents two aluminium products with typical defects. (b) and (c) Visualize the feature maps, in which the color represents the degree of activation. With higher activation, features become more distinguishable. The feature maps obtained by FPN and the proposed AMFF are shown in (b) and (c), respectively.

2) Performance Demonstration: The feature maps of aluminum defects obtained by using FPN and AMFF are shown in Fig. 5. It can be observed that AMFF successfully focuses on defect regions while the background is suppressed. The trends and characteristics of the defects can be accurately

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TABLE VI
PERFORMANCE OF THE SOTA USING AMFF ASSESSED ON ALUMINUM DEFECT DATASET

Detector	Method	mAP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Faster R-CNN [7]	FPN	34.7	48.9	36.0	2.3	4.7	35.6
	FPN *w* / AMFF	**36.1(+1.4)**	**50.9**	**36.7**	**4.6**	**4.2**	**36.9**
	PAFPN	35.2	50.2	36.9	2.4	3.9	36.0
	PAFPN *w* / AMFF*	**36.5(+1.3)**	**50.5**	**38.2**	**3.5**	**4.8**	**37.3**
RetinaNet [9]	FPN	34.1	58.3	32.3	0.3	3.4	36.1
	FPN *w* / AMFF	**35.5(+1.4)**	**61.2**	**33.8**	**2.1**	**2.9**	**37.4**
	PAFPN	34.9	61.5	31.7	3.9	4.8	36.4
	PAFPN *w* / AMFF*	**36.7(+1.8)**	**61.5**	**35.9**	**10.4**	**3.5**	**38.1**
FCOS [44]	FPN	34.8	55.5	34.4	3.4	6.0	35.9
	FPN *w* / AMFF	**37.6(+2.8)**	**60.0**	**36.9**	**5.9**	**6.6**	**38.7**
	PAFPN	37.5	59.9	36.8	4.1	6.3	38.7
	PAFPN *w* / AMFF*	**38.5(+1.0)**	**61.5**	**36.9**	**8.5**	**5.0**	**39.6**

TABLE VII
COMPLEXITY AND REAL-TIME PERFORMANCE COMPARISON OF PREVALENT DETECTORS BEFORE AND AFTER APPLYING THE PROPOSED AMFF. FLOPS AND PARAMETERS (PARAMS) IS PROVIDED TO EVALUATE THE COMPLEXITY, WHILE FPS IS PRESENTED TO EVALUATE REAL-TIME PERFORMANCE

Detector	Method	FLOPs(GFLOPs)	Params(M)	FPS
Faster R-CNN [7]	FPN	282.78	60.16	16
	FPN *w* / AMFF	**400.80**	**76.11**	**14**
	PAFPN	332.35	61.93	15
	PAFPN *w* / AMFF*	**479.37**	**93.83**	**13**
RetinaNet [9]	FPN	282.42	55.28	17
	FPN *w* / AMFF	**390.59**	**71.23**	**16**
	PAFPN	290.32	57.64	16
	PAFPN *w* / AMFF*	**420.10**	**88.36**	**14**
FCOS [44]	FPN	278.42	50.82	18
	FPN *w* / AMFF	**384.60**	**66.77**	**16**
	PAFPN	284.32	53.18	17
	PAFPN *w* / AMFF*	**414.10**	**83.90**	**15**

discerned from the feature heatmaps. Compared with FPN, defect features with AMFF are more easily identifiable.

To further illustrate the effectiveness, AMFF is applied to prevalent detectors equipped with different necks. The results of comparison experiments are presented in Table VI. Compared to FPN, AMFF boosts the performances of detectors by 1.4% mAP, 1.4% mAP, and 2.8% mAP, respectively. Meanwhile, AMFF outperforms PAFPN by 1.3% mAP, 1.8% mAP, and 1.0% mAP, respectively. As anticipated, PAFPN demonstrates superior performance compared to FPN on aluminum defect dataset. AMFF shows superiority over both FPN and PAFPN. Fig. 6 showcases examples of detection results on the aluminum dataset. The results demonstrate the effectiveness of AMFF in detecting weak and small defects.

The conducted experiments and visualizations collectively substantiate the efficacy of the proposed AMFF. The presented results highlight the substantial enhancement of defect features achievable through our approach. Notably, AMFF seamlessly integrates with existing feature fusion networks, resulting in a discernible performance improvement for the detector.

D. Complexity and Real-time Performance Analysis

Network complexity and real-time performance are also the significant concerns to evaluate the practicality of an algorithm. Taking Faster R-CNN with the backbone of ResNet-101 as the baseline, we analyze the floating point operations (FLOPs) and parameters (Params), as well as the frames per second (FPS) to compare the complexities and real-time performances of the methods. As shown in Table VII, with the utilization of AMFF, the FLOPs increase by about 30% while Params by about 25%. As a result, inference speeds

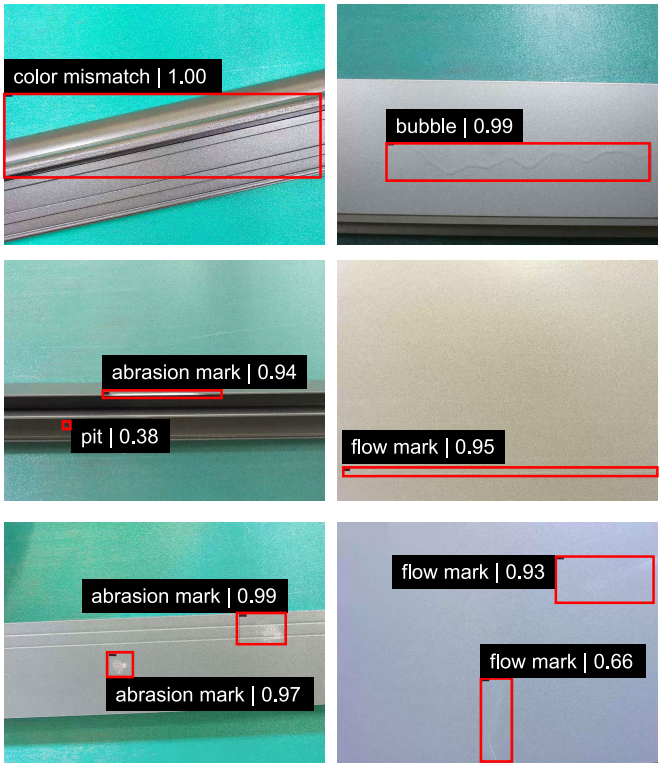


Fig. 6. Examples of detection results on aluminum defect dataset.

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are reduced by 1–2 frames/s. AMFF could meet real-time requirements in industrial applications, and the additional time cost brought by AMFF falls within the range of acceptance.

E. Limitation and Discussion

As demonstrated by Tables I and IV, dilated convolution effectively improves the performance of defect detection by enriching the context information, but at the cost of increasing the computations. Actually, the utilization of dilated convolutions accounts for about 80% of the computation cost of AMFF. With the default backbone network, AMFF still achieves satisfying real-time performance. However, when using a more complicated backbone network, the increased computation cost translates to requests for more computing resources or higher hardware performance, thereby limiting the potential applications of the proposed method. In addition, the proposed AMFF greatly improves the discrimination between defect features and background, and thus improves the accuracy of defect identification and localization. However, AMFF does not emphasize the distinction between various types of defects, leading to defect misclassification. Using the enhanced features to achieve higher classification accuracy not only between defects and background but also between different types of defects needs further investigation.

V. Conclusion

In this article, we incorporate attention mechanism into the feature fusion network to construct the attention-based multiscale feature fusion module. Two integral components of AMFF, namely SEAM and CEAM, are included to perform intralayer and cross-layer attention, respectively. SEAM is conducted on a single feature map that uses multiple parallel dilation convolutions with different rates to enrich contextual information, followed by SA and CA. CEAM performs on multiple feature maps, in which the adjacent feature map is first refined with the guidance of the current feature map and then fused in an additional manner. AMFF improves feature representation as well as identification, and can be seamlessly integrated into different feature fusion networks and applied to detectors for efficient detection. Experiments are conducted on two defect datasets to demonstrate that, AMFF effectively improves the performances of the detectors with acceptable computational cost. The effectiveness and superiority of AMFF indicate the further potential for real defect detection tasks.

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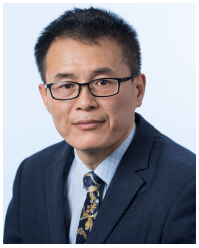
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